AD145A4 Datasheet

Zhuhai Jieli Technology Co.,LTD

Version：V1.2

Date：2024.09.05

ZHUHAI JIELI TECHNOLOGY CO., LTD

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AD145A4 Features

CPU Core

- 32-bit CPU,the highest frequency is 160MHz

Clock Source

- RC Clock frequency about 16MHz
- LRC(low power RC) clock frequency about 32KHz
- HTC(low drift internal high frequency RC)clock frequency is 5MHz

Digital I/O

- 16 programmable digital I/O pins
- USB DP/DM can be configured to normal I/O pins
- General the IO supports pull-up(10k),pull-down(60k), strong,weak output,input and high impedance
- Up to 8 external interrupt/wake-up source(low power available,can be multiplexed to any I/O, with hardware filter)
- Input channel and Output channel, provide arbitrary IO input and output options for some modules

Digital peripherals

- One Full Speed USB 1.1 PHY
- Two UART Controllers(UART0/1) UART1 supports DMA and Flow Control
- Two SPI Controllers with DMA(SPI0/1) support master mode and slave mode.
- One Spi Flash Controller to run code
- Built-in SPI flash
- I2S audio interface
- Two 16-bit Asynchronous Divider Timers
- One IIC Controller
- Four channel PWM output
- 0.5 watt Class-D audio amplifier output
- Infrared remote control decoder
- Watchdog

Analog Peripherals

- MIC amplifier circuit
- Two analog audio input channels
- 10-bit high precision ADC
- 16-bit high precision ADC (mainly as recording)
- 16-bit high precision DAC
- Low voltage protection
- Power on reset

PMU

- Soft off current: <2uA
- Power down current: <30uA
- Music mode: <6mA@HSB 80M
- Recording mode: <11mA@HSB 80M
- LVD range(3bit):1.8V~2.5V, step100mV
- VBAT range : 2.0V~5.5V
- IOVDD range : 2.0V~3.4V

Temperature

- Operating temperature: -40℃ to +85℃
- Storage temperature: -65℃ to +150℃

Package

- QSOP24

Application

- Sound Toy
- Audio player

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1 Block Diagram

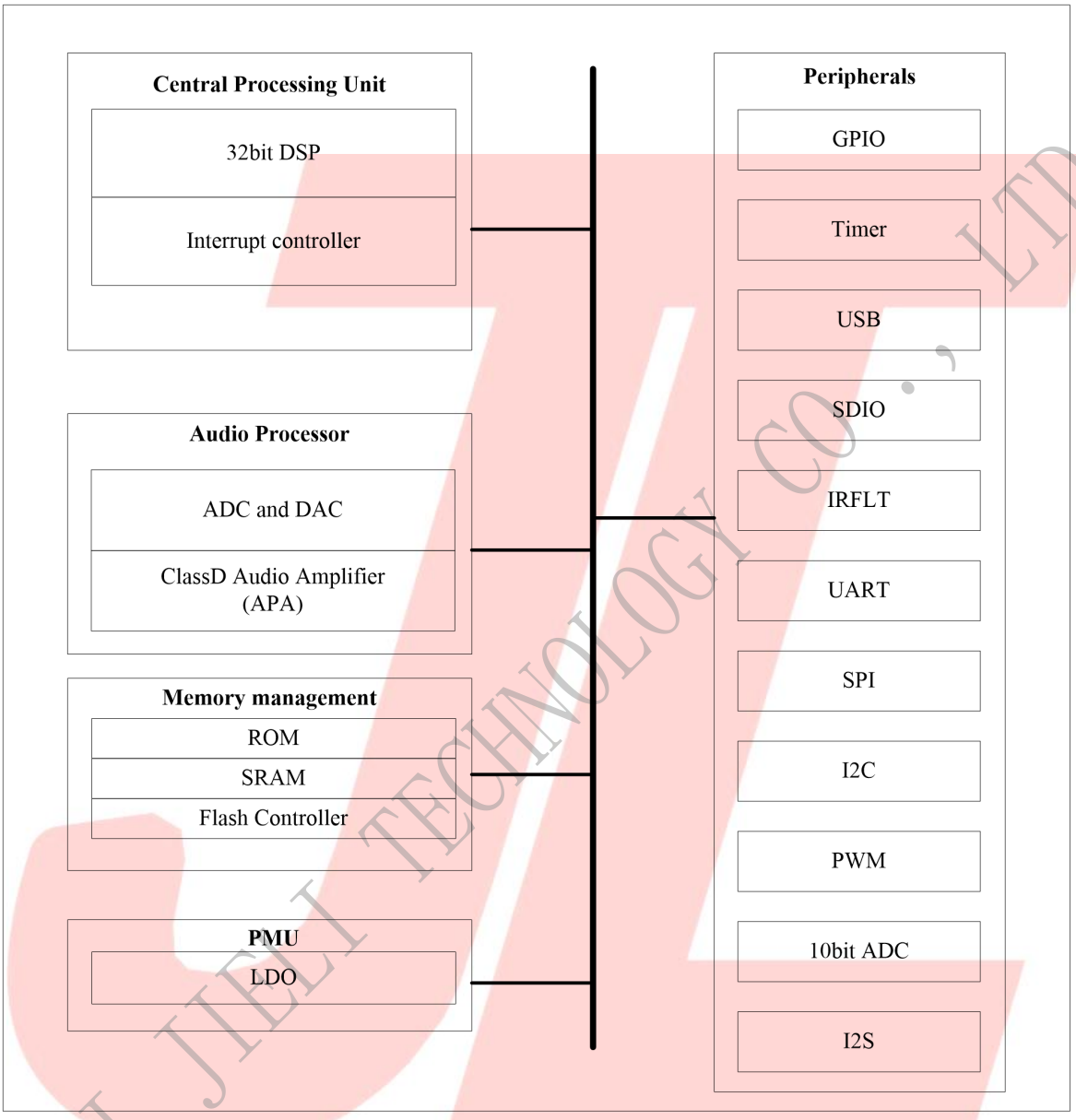

Figure 1-1 AD145A Block Diagram

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2 Pin Definition

2.1 Pin Assignment

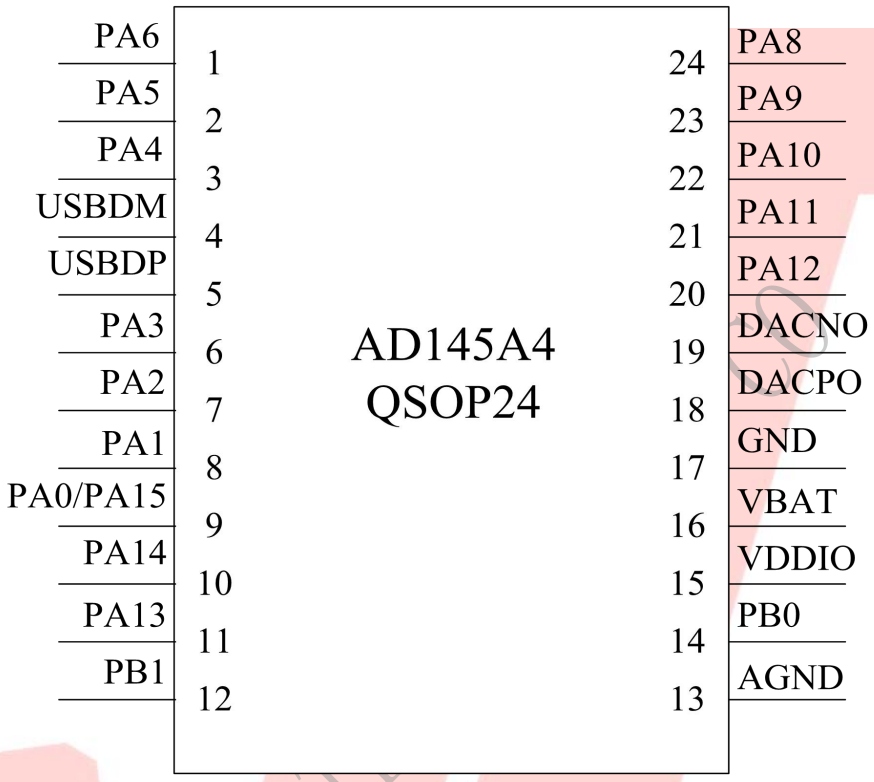

Figure 2-1 AD145A4_QSOP24 Package Diagram

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2.2 Pin Description

Table 2-1 AD145A4_QSOP24 Pin Description

PIN NO.	Name	Type	Drive (mA)	Function	Description
1	PA6	I/O	8/64	GPIO	I2S_MCLK:Audio Link Master Clock; SPI1DIC:SPI1 Data In(C); SD0DATD:SD0 Data(D);
2	PA5	I/O	8/64	GPIO	ADC7:ADC Input Channel 7; SPI0DAT3:SPI0 Data Out3; SPI1DOC:SPI1 Data Out(C); SD0CMDC:SD0 Command(C); SD0CMDD:SD0 Command(D); UART0RXA:Uart0 Data In(A); I2C_SDA(C); PWM1:PWM Channel1 Output;
3	PA4	I/O	8/64	GPIO	ADC6:ADC Input Channel 6; SPI0DAT2:SPI0 Data 2; SPI1CLKC:SPI1 Clock(C); SD0CLKC:SD0 Clock(C); SD0CLKD:SD0 Clock(D); UART0TXA:Uart0 Data Out(A); I2C_SCL(C); TMR2:Timer2 Clock In; PWM0:PWM Channel0 Output;
4	USBDM	I/O	10	USB Negative Data (pull down)	ADC5:ADC Input Channel 5; SPI1DOA:SPI1 Data Out(A); SD0DATC:SD0 Data(C); UART1TXA:Uart1 Data Out(A); I2C_SDA(A);
5	USBDP	I/O	10	USB Positive Data (pull down)	ADC4:ADC Input Channel 4; SPI1CLKA:SPI1 Clock(A); UART1RXA:Uart1 Data In(A); I2C_SCL(A);
6	PA3	I/O	8/64	GPIO	ADC3:ADC Input Channel 3; SPI0DIB(1):SPI0 Data1 In(B); SPI1DIA:SPI1 Data In(A); SD0DATA:SD0 Data(A); PWM2L; MCAP0:Motor Timer0 Capture;

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7	PA2	I/O	8/64	GPIO	ADC2:ADC Input Channel 2; SPI0DOB(0):SPI0 Data0 Out(B); SD0CMDA:SD0 Command(A); I2C_SDA(B); PWM2H;
8	PA1	I/O	8/64	GPIO	ADC1:ADC Input Channel 1; SPI0CLKB:SPI0 Clock(B); SD0CLKA:SD0 Clock(A); UART0RXB:Uart0 Data In(B); I2C_SCL(B); CAP2:Timer2 Capture;
9	PA0	I/O	8/64	GPIO (pull up)	Long Press Reset; ADC0:ADC Input Channel 0; UART0TXB:Uart0 Data Out(B);
	PA15	I/O	8/64	GPIO	ADC12:ADC Input Channel 12; MIC_LDO:Microphone Power Output;
10	PA14	I/O	8/64	GPIO	ADC11:ADC Input Channel 11; AUX1:Analog Channel 1 Input;
11	PA13	I/O	8/64	GPIO	ADC10:ADC Input Channel 10; AUX0:Analog Channel 0 Input; MIC_BIAS:Microphone Bias Output; CAP0:Timer0 Capture;
12	PB1	I/O	8/64	GPIO	MIC_IN:MIC Input Channel;
13	AGND	G	/		Analog Ground;
14	PB0	I/O	8/64	GPIO	**DAC:Analog Audio Output;** ADC13:ADC Input Channel 13; LVD:Low Voltage Detect;
15	VDDIO	P	/		GPIO Power;
16	VBAT	P	/		Battery Power Supply;
17	GND	G	/		Digital Ground;
18	DACPO	O	/		Class-D APA Positive Output;
19	DACNO	O	/		Class-D APA Negative Output;
20	PA12	I/O	8/64	GPIO	I2S_LRCK:Audio Link Word Select; SPI1DOB:SPI1 Data Out(B); SD0CMDB:SD0 Command(B); MCAP3:Motor Timer3 Capture;
21	PA11	I/O	8/64	GPIO	ADC9:ADC Input Channel 9; I2S_SCLK:Audio Link Serial Clock; SPI1CLKB:SPI1 Clock(B); SD0CLKB:SD0 Clock(B); MCAP2:Motor Timer2 Capture;

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22	PA10	I/O	8/64	GPIO	ADC8:ADC Input Channel 8; I2S_DAT3:Audio Link Data3; SPI1DIB:SPI1 Data In(B); SD0DATB:SD0 Data(B); TMR1:Timer1 Clock In; MCAP1:Motor Timer1 Capture;
23	PA9	I/O	8	GPIO (High Voltage Resistance)	I2S_DAT2:Audio Link Data2; UART1TXB:Uart1 Data Out(B); UART1RXB:Uart1 Data In(B); I2C_SDA(D); CAP1:Timer1 Capture; PWM3:PWM Channel3 Output;
24	PA8	I/O	8	GPIO (High Voltage Resistance)	I2S_DAT1:Audio Link Data1; I2C_SCL(D); TMR0:Timer0 Clock In; PWM2:PWM Channel2 Output; OSCI:Crystal Oscillator Input;

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3 Electrical Characteristics

3.1 Absolute Maximum Ratings

Table 3-1

Symbol	Parameter	Min	Max	Unit
Tamb	Ambient Temperature	-40	+85	℃
Tstg	Storage temperature	-65	+150	℃
VBAT	Supply Voltage	-0.3	5.5	V
$V_{VDDIO33}$	3.3V IO Input Voltage	-0.3	3.6	V

Note:The chip can be damaged by any stress in excess of the absolute maximum ratings listed below.

3.2 PMU Characteristics

Table 3-2

Symbol	Parameter	Min	Typ	Max	Unit	Test Conditions
VBAT	Voltage Input	2.0	3.7	5.5	V	_
V_{VDDIO}	Voltage output	2.0	3.0	3.4	V	VBAT = 3.7V, 100mA loading
I_{VDDIO}	Loading current	_	_	100	mA	VBAT=3.7V

3.3 IO Input/Output Electrical Logical Characteristics

Table 3-3

IO input characteristics						
Symbol	**Parameter**	**Min**	**Typ**	**Max**	**Unit**	**Test Conditions**
V_{IL}	Low-Level Input Voltage	-0.3	_	0.3* VDDIO	V	VDDIO = 3.3V
V_{IH}	High-Level Input Voltage	0.7* VDDIO	_	VDDIO+0.3	V	VDDIO = 3.3V
IO output characteristics						
V_{OL}	Low-Level Output Voltage	_	_	0.33	V	VDDIO = 3.3V
V_{OH}	High-Level Output Voltage	2.7	_	_	V	VDDIO = 3.3V

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3.4 Internal Resistor Characteristics

Table 3-4

Port	General Output	High Drive	Internal Pull-Up Resistor	Internal Pull-Down Resistor	Comment
PA0~PA6 PA10~PA15 PB0,PB1	8mA	64mA	10K	60K	1、PA0 default pull up 2、USBDM & USBDP default pull down 3、internal pull-up/pull-down resistance accuracy ±20%
PA8,PA9 (high voltage I/O)	8mA	_	10K	60K	
USBDP	10mA	_	1.5K	15K	
USBDM	10mA	_	180K	15K	

3.5 Analog DAC(PB0) Characteristics

Table 3-5

Parameter	Min	Typ	Max	Unit	Test Conditions
Frequency Response	20	_	16K	Hz	1KHz/0dB 100kohm loading With A-Weighted Filter
THD+N	_	-65	_	dB	
S/N	_	95	_	dB	
Output Swing	_	0.54	_	Vrms	
Dynamic Range	_	92	_	dB	1KHz/-60dB 100kohm loading With A-Weighted Filter
Output Resistance	_	8.3	_	K	_

3.6 ADC Characteristics

Table 3-6

Parameter	Min	Typ	Max	Unit	Test Conditions
Dynamic Range	_	75	_	dB	1KHz/210mVrms line mode :6dB with cap PGAIS=2
S/N	_	79	_	dB	
THD+N	_	-70	_	dB	

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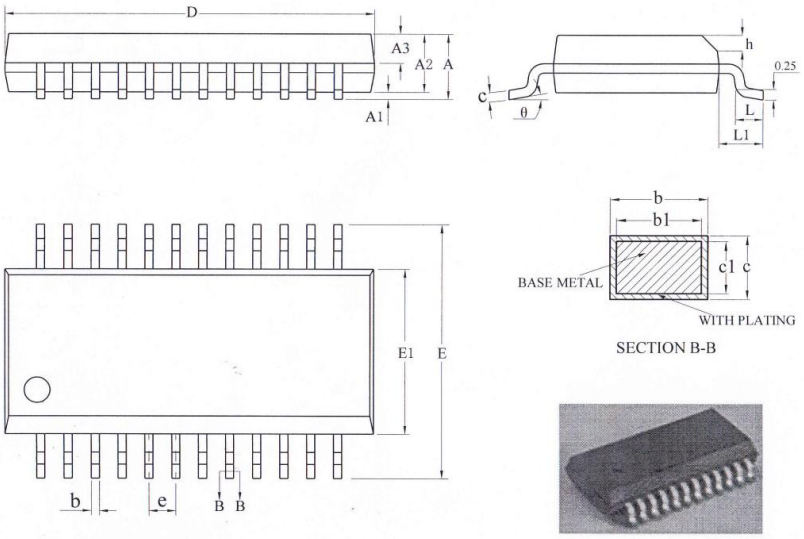
4 Package Information

4.1 QSOP24



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	—	—	1.75
A1	0.10	0.15	0.25
A2	1.30	1.40	1.50
A3	0.60	0.65	0.70
b	0.23	—	0.31
b1	0.22	0.25	0.28
c	0.20	—	0.24
c1	0.19	0.20	0.21
D	8.55	8.65	8.75
E	5.80	6.00	6.20
E1	3.80	3.90	4.00
e	0.635BSC		
h	0.30	—	0.50
L	0.50	—	0.80
L1	1.05REF		
θ	0	—	8°

Figure 4-1. AD145A4_QSOP24 Package

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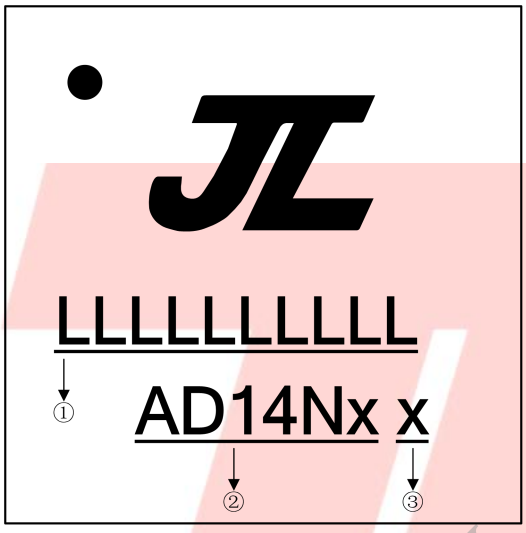
5 IC Marking Information



① LLLLLLLLLL：Production Batch

② AD14Nx: Chip Model

③ x: Built-in flash size

0: No Flash Memory

2: 2Mbit Flash

4: 4Mbit Flash

8: 8Mbit Flash

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6 Storage Condition

6.1 Moisture Sensitivity Level

AD145A is qualified to moisture sensitivity level MSL3 in accordance with JEDEC J-STD-033.

6.2 Storage Alert

1. Calculated shelf life in sealed bag 12 months at＜40°C and 90% relative humidity (RH).
2. Peak package body temperature≤260°C.
3. After bag is opened,devices that will be subjected to reflow solder or other high temperature process must be mounted within 168 hours of factory conditions≤30°C/60%RH or stored per J-STD-033.
4. Devices require bake before mounting if humidity indicator card reads＞10% for level 2a-5a devices or＞60% for level 2 devices when read at 23±5°C,or 3a or 3b are not met.

Please refer to IPC/JEDEC J-STD-033 for baking procedure if necessary.

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7 Revision History

Date	Revision	Description
2021.03.09	V1.0	Initial Release
2023.03.20	V1.1	Modify Features
2024.09.05	V1.2	Add Block Diagram Add IC Marking Information Add Storage Condition Update Features modification.

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